

● Review

FREEHAND 3-D ULTRASOUND IMAGING: A SYSTEMATIC REVIEW

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(Received 4 January 2017; revised 1 June 2017; in final form 5 June 2017)

Abstract—Two-dimensional ultrasound (US) imaging has been successfully used in clinical applications as a low-cost, portable and non-invasive image modality for more than three decades. Recent advances in computer science and technology illustrate the promise of the 3-D US modality as a medical imaging technique that is comparable to other prevalent modalities and that overcomes certain drawbacks of 2-D US. This systematic review covers freehand 3-D US imaging between 1970 and 2017, highlighting the current trends in research fields, the research methods, the main limitations, the leading researchers, standard assessment criteria and clinical applications. Freehand 3-D US systems are more prevalent in the academic environment, whereas in clinical applications and industrial research, most studies have focused on 3-D US transducers and improvement of hardware performance. This topic is still an interesting active area for researchers, and there remain many unsolved problems to be addressed. (E-mail: mmoza102@uottawa.ca) © 2017 World Federation for Ultrasound in Medicine & Biology.

Key Words: Three-dimensional ultrasound imaging, Three-dimensional ultrasound freehand systems, Three-dimensional ultrasound calibration, Three-dimensional ultrasound reconstruction, Three-dimensional ultrasound sensorless methods, Systematic review.

INTRODUCTION

Medical imaging can be traced back to the discovery of X-rays in 1895, and since then, many modalities, such as magnetic resonance imaging (MRI), ultrasound (US), computed tomography (CT), positron emission tomography (PET), medical X-rays and single-photon emission computed tomography (SPECT), have been invented. Ultrasound imaging appears to be the most prevalent in clinics because of its price, tolerability, space, portability and fast performance. Here, common methods for freehand ultrasound imaging in the 3-D domain are systematically reviewed.

BACKGROUND

The 3-D domain for US imaging is a promising and burgeoning approach for visualizing and depicting the inside of the human body in 3-D volumes. The first attempts to use US machines to capture 3-D US volume can be traced back to the 1970s. The Kretz Combison 330, unveiled in 1989, was the first commercial 3-D scanner (Prager et al. 2010). The main advantages of the 3-D

domain for US systems in comparison to 2-D images can be summarized as follows (Fenster and Downey 2000; Fenster et al. 2011; Gebhard et al. 2015; Gee et al. 2003; Hsu et al. 2007; Mercier et al. 2005; Nelson and Pretorius 1998; Solberg et al. 2007; Yu et al. 2011):

- *Visualization of the entire structure of an organ in three dimensions:* The target object and region of interest (ROI) can be visualized as a 3-D surface or 3-D rendered volume by using interpolation, segmentation and registration methods in 2-D or 3-D space.
- *Less dependence on an operator:* The 3-D visualization of anatomy during a diagnostic examination can be screened directly on the user interface instead of mentally transforming 2-D B-scans to perceive a 3-D view. Consequently, it eliminates operator interpretation dependence in the acquisition process. Furthermore, to work with a 2-D US transducer, the operator must be an experienced operator with knowledge of human anatomy, whereas for 3-D US systems, the acquisition step is much easier and less user dependent.
- *Orientation-independent visualization:* Working in the 3-D domain, it is possible to view any arbitrarily oriented planes, even those that are not possible to visualize by conventional US methods.

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Table 1. Quantitative studies of methods in terms of implementation and performance

Reference	Acquisition system	Localization device	Reconstruction method	Visualization software	Accuracy	Performance time	Calibration phantom
Cenni et al. 2016	PC-based US machine + 10.0-MHz linear transducer + 30-Hz transducer HL9.0/60/128 Z, Teemed Echo Blaster 128 Ext-1 Z system	First system: three integrated cameras + OPS (OptiTrack) Second system: OPS (10 optical cameras, Vicon Motion Systems)	VBM	Custom-designed Py3-D FreeHand in Python 2.7	VEA: ~3% VRA: ~1 mm	N/A	Crossed wire embedded in water tank
Daoud et al. 2015	SonixTouch Q+ Ultrasonix + L14-5/38 linear transducer 7.2 MHz	EPS (trakSTAR, NDI)	PBM	VTk + ParaView	N/A	N/A	Double N-wire
Huang et al. 2015	Sonix RP Ultrasonix + linear transducer (L14-5/38) + convex transducer (C3-7/50)	EPS (miniBIRD)	FBM using Bezier curves	Custom-defined	VRA: reduced 0.51–5.07%	Reconstruction in ~20 s and scanning in ~1 min	Fetus CIRS
Wen et al. 2015	DC-7, Mindray Medical International	OPS (Polaris)	VBM using Bayesian-based non-local method	Custom-designed	VEA: ~8%	Reconstruction: ~2142 s	Abdominal CIRS Modal 057
Chen et al. 2014	2-D US device DC-7 Mindray	EPS (Aurora)	VBM using kernel regression model	Custom-designed	RMSE: 6.68	Reconstruction: ~380 min	Standard abdominal (CIRS)
Huang et al. 2013	Sonix RP, Ultrasonix + linear transducer (L14-5/38) + convex transducer (C3-7/50)	MPS (linear sliding track + Digital Scale Units, Model 812-103 + Bluetooth module)	VBM using squared distance-weighted interpolation	VTk + Custom-designed on C++	VRA: improved by 0.46%–2.14% VEA: ~1.46%	Reconstruction: 251 s for 793 B-scans	CIRS Model 044
Neshat et al. 2013	Toshiba PVT-375 BT, 1.9–6 MHz + Aplio XG SSA-790 A + handheld mechanical scanning device	EMPS	PBM	N/A	VRA: ~1.6 mm	Scanning time 6–10 s	String embedded 15% glycerol/distilled water solution
Wen et al. 2013	DC-7 Mindray Medical International Ltd	EPS Aurora	PBM with fast marching interpolation	Custom-designed with C++	N/A	220 * 202 * 145 voxels can be reconstructed within 30 s	Abdominal CIRS
Toonkum et al. 2011	N/A	MPS linear tracking	VBM using cyclic regularized Savitzky–Golay filters	Custom-designed in MATLAB	NMSE: 0.0321	Reconstruction: ~600 s	N/A
Qiu et al. 2011	LANDWIND ultrasound scanner	OPS (NDI)	VBM using improved distance-weighted interpolation	VTk	N/A	Reconstruction: 254 s with four neighbors	Water based with chicken kidney
Yu et al. 2011	Acuson Sequoia C256 + 7 MHz array transducer 7 V3 C	EPS (EPOM Flock of Birds)	VBRM using multiview visualization	Custom-designed in C and MATLAB	VEA: ~5%	N/A	CIRS Model 055
Scheipers et al. 2010	Ultrasonix + L12-5 linear transducer at 10 MHz	Custom-made OPS (Polaris)	Direct frame interpolation VBM using spherical linear interpolation of quaternions	Custom-designed in MATLAB	VEA: ~25 %	Reconstruction: ~100 s	N/A
Dewi et al. 2009	GE LogiQ 9 ultrasound scanner (GE Healthcare) + 2-D transducer linear matrix array 10 L 6.3–10 MHz	EPS (pier, Ascension Tech) + 3-D Free Scan (Echotech 3-D Imaging Systems)	PBM using improved Olympic method	MATLAB 7.0.4	MAE: 0.0069	N/A	N/A
MacGillivray et al. 2009	5- to 10-MHz linear ultrasound transducer (Diasus)	OPS (Polaris)	VBM	Stradwin + ANALYZE	VEA: ~16% of corresponding MR derived	N/A	N/A
Huang and Zheng 2008	Portable US (SonoSite 180 PLUS)	EPS (miniBIRD)	VBM using median filters	Custom-designed on C++	N/A	Reconstruction: ~45 min with 1 GB RAM + 2.8 Ghz CPU	Cross-wire phantom

Goldsmith et al. 2008	Terason 3000 + linear array transducer 5–9 MHz	OPS + three-axis MEMS gyroscope	VBM	Stradwin	MAE: 3.84%	N/A	CIRS Model 044
Coupe et al. 2007	Sonosite system + 4- to 7-MHz transducer + for intra-operative sequences, the sinuosity cranial transducer + Sononav Medtronic system as OPS	EPS (miniBIRD)	VBM using probe trajectory interpolation	Stradx + StackX + StackSX	RMSE: 0.381 mm MSE: ~25	Reconstruction: ~ 120 s	CIRS
Poon and Rohling 2006	Voluson 730 Pro + 4- to 8-MHz curved-array 3-D US transducer	OPS (Optotrak 3020)	M/I: PBM + linear interpolation + VBRM	3-D View 2000 + VTK	VRA: ~0.4 mm VEA: ~0.4%	Re-construction: ~40 min	Fetus phantom (CIRS)
Yu et al. 2006b	Acuson Sequoia C256 + 7-MHz 1-D wide-view array transducer 7 V3 C + ECG gating	DC EMP, the Flock of Birds (FOB)	VBRM using a 3-D Hotelling transform	N/A	N/A	N/A	CIRS Model 055
Huang et al. 2009a; Huang and Zheng 2005	Portable US (SonoSite 180 PLUS)	EPS (miniBIRD)	VBM using adaptive distance-weighted method	Custom-designed C++	N/A	Re-construction: ~10 min	Cross-wire
Huang et al. 2005	Portable ultrasound SonoSite 180 Plus + linear transducer (L38/10-5 MHz)	EPS (miniBIRD)	VBM using sphere region	Custom-designed VTK using ray casting	VRA: ~0.5 mm VEA: ~5.44 %	N/A	Cross-wire using two cotton wires in water tank
Kohyama et al. 2005	SonoSite180 PLUS + linear array L38 transducer 5.32 MHz	OPS (Polaris)	FBM using marching cubes method	N/A	N/A	N/A	Cross-wire
San José-Estépar et al. 2003	Hitachi EUB 515 images were digitized in RGB PAL format at 25 frames/s	EPS	Weighted sum of the voxels in neighborhood using spherical Gaussian kernel (VBM)	Stradx + VTK	N/A	N/A	Single-wall
Treece et al. 2003	Diasus US machine 1, using 5- to 10-MHz + 10- to 22-MHz linear-array transducers	OPS (Polaris 2) AdapTrax3	VBM using Stradx	Stradx	VRA: ~0.5 mm	N/A	Nylon wires embedded
Gobbi and Peters 2002	Aloka SSD-1700 scanner + 5.0 MHz curved-array neuro-ultrasound transducer	OPS (POLARIS) Northern Digital	Splatting technique with several kernels + boxcar splat as pixel trilinear interpolation (PBM)	VTK on C++	VRA: 0.4 mm	N/A	Deformable PVA-cryogel
(Zhang et al. 2002, 2004)	Ultrasonix	OPS (OPTOTRAK2 3020)	FBM using radial basis functions	N/A	VRA: ~1 mm	Re-construction within 1 min with 1-GHz CPU and 1 GB RAM	Z-wire
Treece et al. 2001a	Toshiba Powervision 7000 + 3.75 MHz convex curvilinear array transducer	EPS (Polhemus FASTRAK)	RBM	Stradx	VRA: ~ 1 mm VEA: 2% <i>in vitro</i> , 5% <i>in vivo</i>	10 s for reslicing	N/A
Barratt et al. 2001a	HDI 5000, ATL-Philips + L12-5 transducer (5- to 12-MHz linear-array transducer) + ECG grabber	EPS (EPOM pcBIRD)	N/A	Custom-designed software	VRA: ~1.30 mm VEA: ~3.39%	N/A	Water-based with upright pin embedded in a glued rubber bung
Krücker et al. 2000	Logiq 700 (GE Medical Systems) + M12, 1.5 D matrix array transducer in frequency of 11 MHz	MPS (elevational linear position using a potentiometer + rotational orientation using an optical encoder)	VBRM	MIAMI Fuse software	ARE: ~0.5 mm	Typical volume: 5–40 s	Densely packed, random distribution of fine particles, spherical void and two nylon filaments
Poulsen et al. 2005	PC-based system Terason	OPS Agilent ADNS-2610	N/A	Sonocubic	VEA: 6%	N/A	Pure agar rod embedded

(Continued)

Table 1. Quantitative studies of methods in terms of implementation and performance (*Continued*)

Reference	Acquisition system	Localization device	Reconstruction method	Visualization software	Accuracy	Performance time	Calibration phantom
GP006-0047 (Trece et al. 1999b)	Toshiba Powervision 7000a + 3.75-MHz convex curvilinear array transducer	EPS (Polhemus FASTRAK)	VBM	Stradx	VRA: ~1 mm VEA: 5% for kidney, 7% for bladder VEA: ~2%	Whole process about 45 s	M/I
Gilja et al. 1998	Diasonics V. VST Master Series + 2.25 MHz phased array and 3.5-MHz curved array transducer	EPS (Flock of Birds)	VBRM	ViewPoint	N/A	N/A	Water-based embedded pig skin, fat and muscle layers
Barry et al. 1997	Toshiba SSH 140a, 7.5-MHz + linear array transducer at 25 fps + ECG signal grabber	EPS (EPOS) Polhemus 3 Space Isotrak II	Grid mapping (PBM)	TOSCA	VRA: 1 mm, VEA: 1.1%	<30 min for 6000 images JPEG 15:1 and 75% quality occupying	Water-filled balloon using galactose solutions

VEA = volume estimation accuracy (deviation from the actual volume); VRA = volume reconstruction accuracy; EPS = electromagnetic positioning system; OPS = optical positioning system; EMPS = electromechanical positioning system; MPS = mechanical positioning system; TOSCA = tools for segmentation; correlation and analysis; VTK = Visualization Toolkit; N/A = not available; M/I = many items; VBM = voxel-based method; PBM = pixel-based method; RBM = function-based method; VBRM = function-based method; ARE = average registration error; MAE = mean absolute error; VBRM = volume based on registration method; RBM = reslicing-based method; RMSE = root mean square error; NMSE = normalized mean square error; MSE = mean square error.

The 3-D methodology also enables clinicians to discuss, re-examine and depict the inside of a patient's body from different viewpoints after they have left the clinic, and it is also possible to share those data with other experts for better diagnosis by finding undetected disease features in the data.

- *Measurement of quantitative attributes in three dimensions:* In many clinical applications, the physicians must measure the volume or other quantitative properties of the targeted organ, and it is more likely that employing 3-D volumes for measurement is more accurate than using only 2-D images.
- *Repeatability of ROI examination:* To monitor the progression of therapy, to perform a follow-up study of a disease, to observe the response of the body to the therapy and to plan treatment, physicians like to see the ROI from exactly the same perspective as in the previous examination. In conventional 2-D US imaging, accomplishing this task is arduous because of misregistration and difficulties in adjusting the transducer into a similar position. Use of 3-D US volumes significantly facilitates these types of assessments and improves the accuracy of the results.
- *Fusion with other 3-D image modalities:* Multimodal registration of 3-D blocks of US data can be combined with the output results of other imaging modalities, such as MRI, which is more efficient and valuable than the case of image fusion in the 2-D domain.

The freehand method has advantages over use of a 3-D US transducer, such as lower cost, higher volume quality and improved field of view (FOV) (Fenster and Downey 2000; Mercier et al. 2005). However, capturing a 3-D US volume with the freehand technique is difficult and sometimes cumbersome because of errors from the localization sensors, low accuracy of calibration, the many delays at each step of the reconstruction algorithm (Gee et al. 2003; Mercier et al. 2005) and the use of high-performance computing systems such as graphical processing unit-based visualization (Dai et al. 2010; Gobbi and Peters 2002) (a list of computing system setups can be found in Table 1). Therefore, many researchers have attempted to overcome the difficulties of freehand methods.

OBJECTIVES

The aim of this systematic review was to investigate 3-D US imaging comprehensively with a focus on the freehand 3-D US and reconstruction techniques that have been used in academic studies and medical applications. Ultimately, the main questions are addressed in this review are as follows:

- What are the current situation and trends of freehand 3-D US imaging in research fields?
- What methods have been used for freehand 3-D US imaging thus far?
- What are the main limitations, and what issues or difficulties are involved?
- What are the standard criteria used for comparing freehand 3-D US imaging methods?
- Which clinical applications have been addressed using freehand 3-D US imaging methods?
- Which researchers, research groups, publications and countries are leading on this topic?

A classification of 3-D US systems

Thus far, many methods and systems for constructing a 3-D US volume have been designed, implemented and tested. The main difference between these systems can be seen in the acquisition procedures. A taxonomy of 3-D US systems in terms of scanning methods can be considered in the categories in the following subsections (Fenster and Downey 2000):

3-D US transducers

Ultrasound transducers have been designed so that they can capture 3-D volumes directly using a 2-D array of piezoelectric materials (Bureau et al. 1998; Dhanantwari et al. 2004; Krücker et al. 2000; Light et al. 1998; Prager et al. 1998a; Steiner et al. 1994; Yeh et al. 2006) or using a mechanical or electrical controllable 1-D array of piezoelectric materials embedded within a probe housing that sweeps 3-D space in each cycle (Mercier et al. 2005). As an alternative technique, many researchers have tried innovative methods such as using two additional perpendicular 1-D arrays of piezoelectric materials added to the main array for position tracking (Hossack et al. 2000). Another example is a ring array of piezoelectric materials for use in intravascular applications (Yeh et al. 2006).

In general, the use of 3-D probes results in worse quality than the use of 2-D probes (Mercier et al. 2005). The 3-D probes tend to be larger and more expensive as well. The main difficulties in using 3-D US probes to acquire a proper volume as follows: optimum settings of the US system, selection of beamforming method, filtering and timing techniques and limitations on the number of cells in the 2-D array (see Fig. 1b).

Freehand 3-D US imaging

In these approaches, an operator moves a tracking device that is attached to a 2-D US transducer on the ROI; then, acquired data (B-scans along with their corresponding positions and orientations) are sent to a computer for further processing and volume reconstruction.

The tracking system can consist of one of the following or a combination (Dewi et al. 2015; Tatar 2006): electromagnetic sensors, acoustic or optical sensors, mechanical arms, fiber-optic cables, inertial measurement units (IMUs), skin feature tracking using cameras or laser sensors and any other systems that can track the US transducer position and orientation accurately with six degrees of freedom (6 DOF). In freehand systems, electromagnetic sensors are the most cost-effective and convenient method for localization. Two types of magnetic fields are used for electromagnetic sensors: AC or DC (Pagoulatos et al. 2000). DC magnetic sensors are less susceptible to metallic materials, whereas AC types are more accurate. Electromagnetic and optical sensors have been used in many research studies because of their high precision compared with other sensors; their main drawbacks are metal sensitivity and line of sight, respectively.

Because a location tracking system specifies only the positions and orientations of the US transducer itself rather than the image output from the US probe, spatial calibration is mandatory before translating each pixel of the B-scans to the corresponding voxel in the 3-D volume. Furthermore, for synchronization of the B-scans and location data, a temporal calibration and a filtering step, in which redundant data are removed, must be conducted.

Standard linear ultrasound transducers are usually used in this technique. However, the freehand method is also used for clinical applications such as bronchoscopy, endoluminal, endoscopy and endosonography (Table 2) using special types of US transducers, which are connected to a tracking system (Fig. 1a,d,e).

Using a 2-D scanner with an external fixture

In this approach, a 2-D US transducer, usually in transrectal ultrasound (TRUS), is moved in a pre-defined direction by an external fixture. This fixture is connected to a mechanical or electromechanical device that moves the probe in a linear, rotational or tilting path near the ROI. In each scan, the position and orientation of the probe are known and are used for reconstruction. This type of 3-D US imaging is usually employed in clinical applications such as endorectal procedures (mainly for prostate brachytherapy), cancer biopsy and needle pathfinding (Bax et al. 2008; Wei et al. 2004; Yan et al. 2012) (Fig. 1c).

Sensorless methods

Theoretically, it is possible to reconstruct 3-D US volumes without using any external location tracking device. In these techniques, the relationship between the statistical information on the speckles in different images, such as linear regression, frame distance

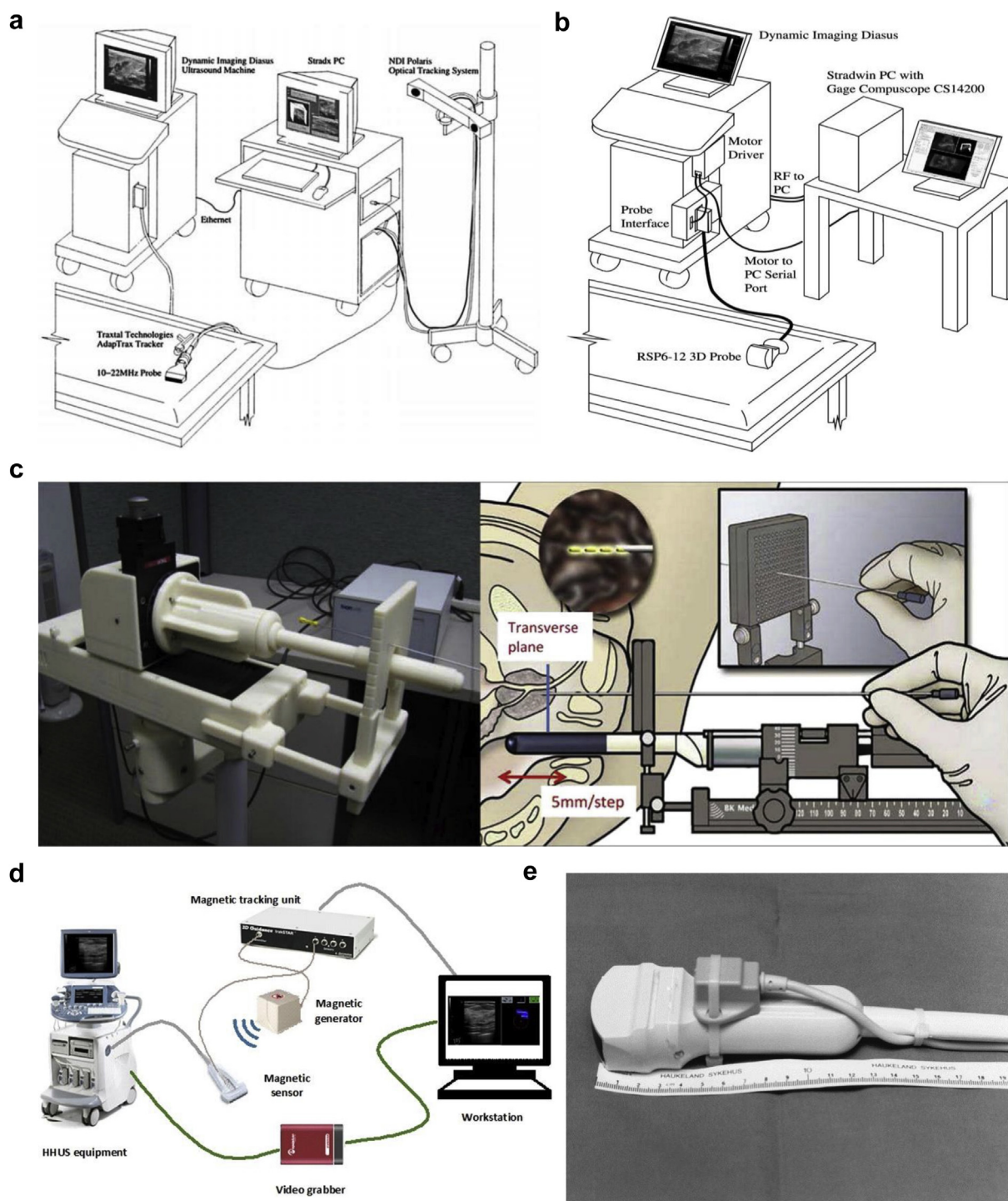


Fig. 1. Some of the 3-D ultrasound imaging and reconstruction systems. (a) Freehand method using an optical tracking system (Treece et al. 2003). (b) Freehand method using a 3-D ultrasound transducer (Treece et al. 2008). (c) Freehand method using a mechanical external fixture (Yan et al. 2012). (d,e) Freehand method using an electromagnetic tracking system (Gilja et al. 1998; Huang et al. 2017).

estimation and decorrelation, is used for localization of the images. This method of 3-D US construction is an innovative and promising alternative for the future of 3-D US imaging, although the accuracy of these

methods is still not comparable to that of other approaches (Li et al. 2002).

In freehand or sensorless methods, the data (B-scans, positions and orientations) are captured after

Table 2. Clinical applications of freehand 3-D ultrasound imaging

Application or organ or organ phantom	References
Carotid arteries	Barratt et al. 2001a, 2001b; Barry et al. 1997; Chung et al. 2016; Fenster and Downey 2003; Fenster et al. 1998; Landry et al. 2005; Pelz et al. 2017; Picot et al. 1993; Prager et al. 1998b; Rosenfield et al. 1991; Roxborough and Nielson 2000; Toonkum et al. 2011; Treece et al. 2001b; Yeom et al. 2014
Coronary arteries	Chatzizisis et al. 2006; Di Mario et al. 1995; Gil et al. 1996; Prause et al. 1997; Roelandt et al. 1994; Rosenfield et al. 1991
Intravascular diseases	Arbab-Zadeh et al. 1999; Chandrasekaran et al. 1994; Ennis et al. 1993; Evans et al. 1996; Klein et al. 1992; Lengyel et al. 1995; Maurincomme et al. 1992; Meairs et al. 2000; Ng et al. 1994; Prati et al. 1996; Reid et al. 1995; Rosenfield et al. 1992; Shekhar et al. 1996; Thrush et al. 1997; Von Birgelen et al. 1997; Wahle et al. 1999a, 1999b; Welsh et al. 2001
Brain tumor, brain structure and neurosurgery	Coupe et al. 2007; Gobbi and Peters 2002; Gronningsaeter et al. 2000; Hartov et al. 1999; Lindner et al. 2005; Lindseth et al. 2002; Mercier et al. 2011; Miller et al. 2012; Penney et al. 2004; Smith et al. 2004
Laparoscopic with augmented reality visualization	Fristrup et al. 2004; Konishi et al. 2007; Nakamoto et al. 2002, 2007, 2008; Sato et al. 2001
Thyroid	Bogush and Tuzikov 2005; Estépar et al. 2003; Prager et al. 1999; San José-Estépar et al. 2003; Schlögl et al. 2004; Treece et al. 2001b
Kidney	Estépar et al. 2003; Ghani et al. 2008; Hughes et al. 1996; San José-Estépar et al. 2003; Strømmen et al. 2004; Treece et al. 1999a
Obstetrics and gynecology (fetus)	Blaas et al. 1998; Carr et al. 2000; Edwards et al. 1998; Fenster and Downey 2003; Gee et al. 2002, 2003, 2004; Gooding et al. 2008; Guerra et al. 2000; Hafizah et al. 2010; Herberg et al. 2005; Lees 2001; Nelson and Elvins 1993; Nelson and Pretorius 1992, 1997; Prager et al. 2002, 2003; Ren et al. 2010; Sakas and Walter 1995; San José-Estépar et al. 2003; Steen and Olstad 1994; Steiner et al. 1994; Timor-Tritsch and Platt 2002; Wee et al. 2011; Yagel et al. 2007
Liver	Beller et al. 2007; Blackall et al. 2005; Boctor et al. 2003; Chen et al. 2014; Coupe et al. 2007; Hughes et al. 1996; Karamalis et al. 2009; Nakamoto et al. 2007; Neshat et al. 2013; Ni et al. 2009; Penney et al. 2004; San José-Estépar et al. 2003; Treece et al. 2001a; Wein et al. 2006; Wen et al. 2015
Neck (jugular vessels) and scapula	Scheipers et al. 2010; Sun et al. 2014; Wein et al. 2006; Worobey et al. 2014
Urinary bladder	Boctor et al. 2001; Pauletzki et al. 1996; Rohling et al. 1999b; Sanches and Marques 2000, 2002; Treece et al. 1999a, 1999b; Wen et al. 2013
Forearm, ankle, ligament, articular cartilage, Achilles tendon, fingers	Cash et al. 2005; Farris et al. 2013; Huang and Zheng 2005; Huang et al. 2005, 2006b, 2009a, 2013, 2015; Lefebvre et al. 1998; Obst et al. 2013, 2017; Prager et al. 1999; Singh et al. 2014; Solberg et al. 2011; Treece et al. 2001b; Weller et al. 2007
Heart	Belohlavek et al., 1991; Berg et al. 1999; Elen et al. 2008; Fulton et al. 1994; Gao et al. 2009; Geisler et al. 2007; Guerra et al. 2000; King et al. 1990; Koolwal et al. 2009, 2011; Kuo et al. 2007; Legget et al. 1998; Pace et al. 2009; Stetten et al. 1998; Treece and Prager 2002; Varandas et al. 2004; Yagel et al. 2007; Yu et al. 2006b
Femur, pelvis, hip joint and their relevant arteries	Barratt et al. 2008; Hirschmann et al. 2011; Lempereur et al. 2013; MacGillivray et al. 2009; Ohbuchi et al. 1992; Passmore et al. 2016; Passmore and Sangeux 2016; Peters et al. 2010; Prager et al. 1999; Sun et al. 2013; Torres et al. 2012
Prostate	Cool et al. 2008; Crivianu-Gaita and Holban 1997; Holmes et al. 2003; Tong et al. 1996
Breast	Krücker et al. 2000; Sato et al. 1998; Xiao et al. 2002
Spine and scoliosis	Cheung et al. 2015; Dewi et al. 2009; Kohyama et al. 2005; Lang et al. 2012; Purnama et al. 2010; Vo et al. 2015
Hepatic system	Boctor et al. 2001; Treece et al. 1999b
Biopsy and needle guidance	Brambati et al. 1987; Cool et al. 2008; Gebhard et al. 2015; Zhang et al. 2006
Medial gastrocnemius and semitendinosus muscle	Barber et al. 2009; Haberfehlner et al. 2016
Eyes	Downey et al. 1996
Elastography	Lindop et al. 2006; Treece et al. 2008
Bronchoscopy, endoluminal, endorectal, endoscopic, and endosonographic	Gravante and Giordano 2008; Ivanov and Diacov 1997; Sumiyama et al. 2003; Tsutsui et al. 2005; Warmath et al. 2004, 2005; Zang et al. 2014

a system calibration, the 3-D US volume or surface is reconstructed and the final results are rendered, resliced and visualized to the desired view. To clarify this procedure, a general overview of the steps in a typical freehand system and sensorless method is provided in [Figure 2](#).

Three-dimensional US reconstruction algorithms are categorized into function-based, voxel-based, pixel-based and hybrid methods according to ordinary classifi-

cations in the literature ([Solberg et al. 2007](#)). Before or after the reconstruction section, the quality of the B-scans is enhanced by applying filters, cropping the ROI and delineating and registering the ROI in the 2-D or 3-D domain for use in the visualization phase.

Many researchers have surveyed 3-D freehand US imaging systems and relevant topics ([Candiani 1998](#); [Fenster et al. 2011](#); [Fenster and Downey 1996, 2000](#);

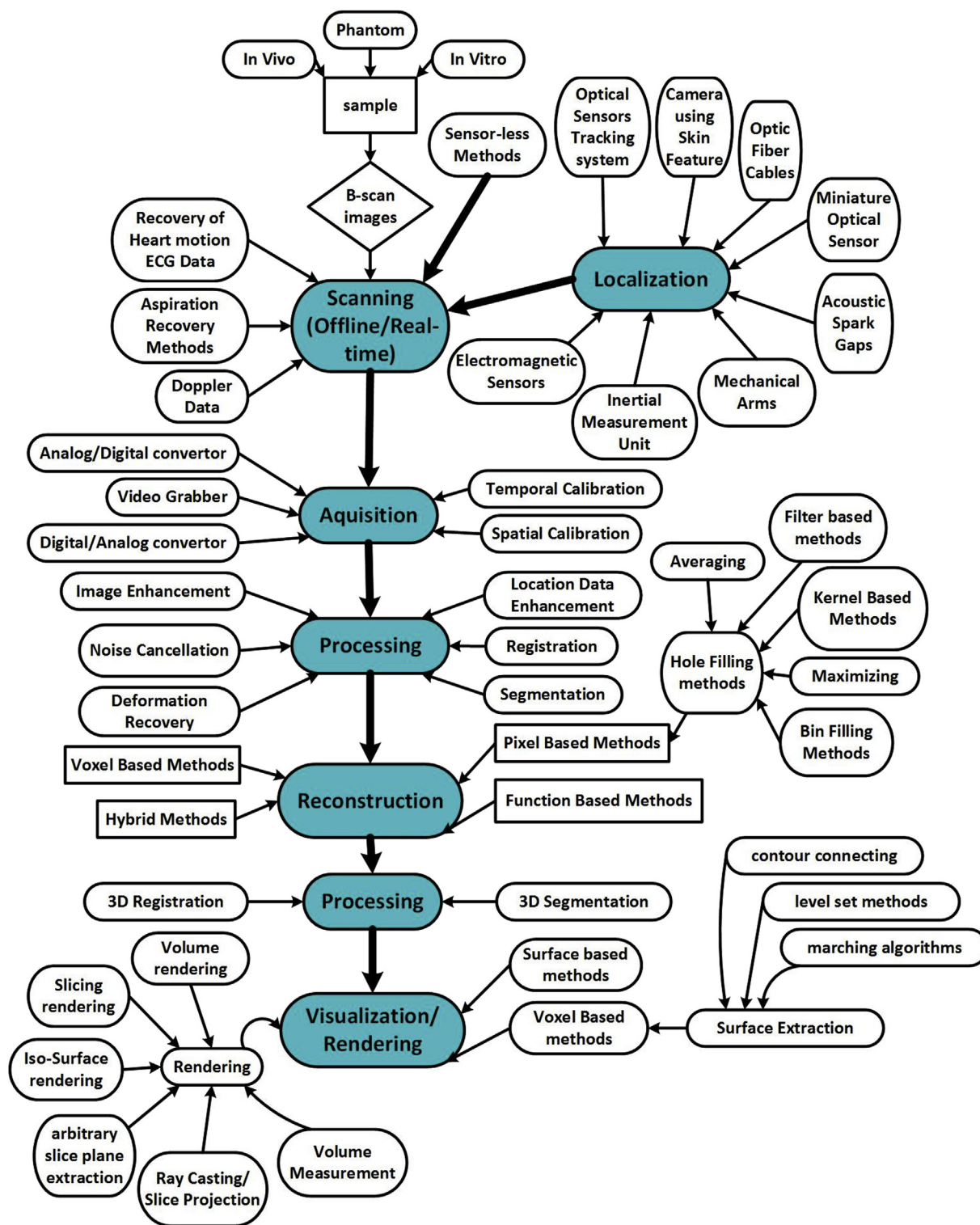


Fig. 2. Overview of freehand and sensorless 3-D ultrasound reconstruction systems.

Gebhard et al. 2015; Gee et al. 2003; Hsu et al. 2007; Mercier et al. 2005; Nelson and Pretorius 1998; Prager et al. 2010; Scheipers et al. 2010; Solberg et al. 2007;

Treece et al. 2003; Yu et al. 2011), and this article is the first systematic review of this topic that covers the literature between 1970 and 2017.

SEARCHING THE MATERIALS AND METHODS

The Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) statement (Liberati et al. 2009; Moher et al. 2009) is a popular systematic review guideline in various fields of medicine, and was used as the basis for reporting a systematic review in this study. We made a few modifications to the PRISMA protocols to be compatible with our study which is more in the engineering domain. We also used other guidelines such as that of Budgen and Brereton (2006) in the computer science area.

We collected and surveyed the relevant papers from 1970 to 2017. For the beginning of our investigation, we chose 1970, the year when the first studies of 3-D US imaging were published. An electronic literature search was conducted between April 2016 and March 2017 using four self-contained and comprehensive databases, which are listed in Table 3. Although each of these databases contains all of the journals, books and conferences related to the subject, we also searched more closely some of the main and popular journals in the literature (Table 4). This search was supplemented by reference lists from dissertation reports, review articles and gray literature searches across the Internet.

We designed a protocol for our systematic review procedure that consisted of planning, conducting and reporting. In the conducting phase, the key words depicted in Figure 3 were employed for the search in the databases.

Eligibility criteria for inclusion and exclusion

Under our designed systematic review protocol, the output result of a selected reference must be a 3-D US volume that was acquired by any class of acquisition, reconstruction and visualization methods in the literature or a novel procedure for 3-D US imaging. We focused more on references that explained a system of freehand 3-D US imaging and applied the method medically or compared it with similar methods. We explored any innovation in 3-D US acquisition techniques, reconstruction algorithms, visualization methods and relevant studies.

References in which the focus was on only one step of the 3-D US imaging procedure were excluded in the first search. For example, references that discussed only the calibration method of freehand or merely described the usage of freehand 3-D US imaging in clinical applica-

Table 4. Main journals used as references for the search

Name of Journal
<i>Ultrasound in Medicine and Biology</i>
<i>Medical Image Analysis</i>
<i>Journal of Medical Ultrasound</i>
<i>Journal of Medical Ultrasound</i>
<i>Ultrasonics</i>
<i>Ultrasound in Obstetrics and Gynecology</i>
<i>Medical Physics</i>
<i>Physics in Medicine and Biology</i>
<i>IEEE Transactions on Medical Imaging</i>

tions without explaining methodology would have been excluded from our investigation. We also excluded relevant dissertations and some book chapters because we could find their results and abstracts in their associated articles. Many studies, especially conference articles, attempted only to explain the general idea of their methods briefly without explaining the details of the procedures and focused the remainder of the article on an illustration of their results. These articles and many relevant hardware complicated patent descriptions were also excluded in the first search phase.

Article selection and analysis

To collect references from each database, we designed and deployed a coding system composed of two letters and seven numbers. Each code indicated the name of the reference database, the page number and the article number in our database. For example, GP002-0017 indicates that the reference is the 17th article in our reference inventory and was found and collected on page 2 in the Google Scholar database using our key words. We used those codes instead of long reference names for simplification in our communications, analysis and referencing. The title, author name(s), country in which the research was done, publication place and year, number of citations on the day of collection, reference abstract, authors affiliations, identification code in our database, reference type (journal, conference, thesis or book chapter), grade number from our assessment, reason for inclusion or exclusion and a brief note about the reference were collected in an Excel spreadsheet. This review is not a meta-analysis and does not include a statistical analysis.

A total of 319 references were retrieved from our initial search of the databases, especially Google Scholar and IEEE Explorer. Ultimately, by searching other databases and hand-searching references, ($n = 68$), we added articles to our repository list. Then, all titles, abstracts, and discussion and conclusion sections were reviewed, and a brief note was written for each. Instead of excluding papers after this phase, we designed a grading system for

Table 3. Reference databases used for the search

Name of reference databases
Google scholar
University of Ottawa library search engine
IEEE Explore
PubMed

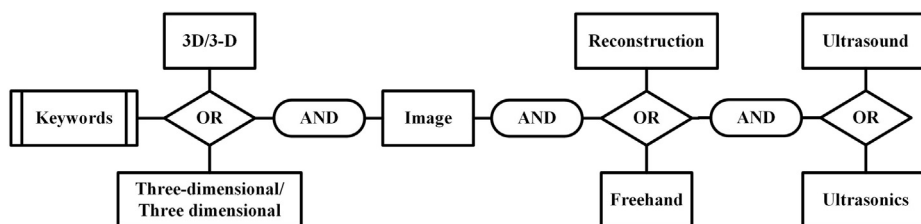


Fig. 3. Key words used in the reference search.

references with an instruction from our protocol (Table 5). Thus, after the skim reading stage, each reference had a grade describing its relevance to the topic. The numbers of references with each final grade (between 0 and 6) are listed in Table 6.

We excluded more references with grades ≤ 3 as non-relevant papers ($n = 154$), and we filtered them in an Excel file. We then excluded ($n = 123$) references with a grade of 4 because we realized that the grade 4 papers were relevant to the topic, but described their methods less clearly or focused on more than one phase of the problem without explaining the other phases of the 3-D US reconstruction. In total, our references were divided into 6 book chapters, 91 conference papers, 284 journal papers, 6 theses/dissertations and 2 patents. The results of the reference collection, inclusion and exclusion steps are illustrated in Figure 4, in which a modified version of the PRISMA flowchart is used.

We then read the full texts of the remaining 94 references and excluded 49 of them because they repeated other methods and provided only small contributions, attempted to work out an old issue that is not a problem today or repeated review papers. For example, many studies used transrectal or transvaginal transducers for 3-D US imaging, did not provide a detailed explanation about their method of 3-D US reconstruction, but only the 3-D volume results. Short analyses and descriptions of the remaining 45 papers (Boctor et al. 2001; Chen

et al. 2009, 2014; Coupe et al. 2007; Deng et al. 2012; Fenster et al. 2011; Gee et al. 2002, 2004; Gilliam et al. 2006; Hassenpflug et al. 2005; Housden et al. 2006a, 2007; Huang and Zheng 2006a, 2008; Huang et al. 2005, 2009b, 2013, 2015; Karamalis et al. 2009; Kohyama et al. 2005; Lindseth et al. 2003; MacGillivray et al. 2009; Pagoulatos et al. 2000; Penney et al. 2004; Poon and Rohling 2006; Prager et al. 2002, 1999, 2003; Qiu et al. 2011; Roxborough and Nielson 2000; San José-Estépar et al. 2003; Sanches and Marques 2003; Scheipers et al. 2010; Sun and Anthony 2012; Sun et al. 2013, 2014; Toonkum et al. 2011; Treece et al. 2001a; Varandas et al. 2004; Wen et al. 2013, 2015; Yu et al. 2006a, 2005; Zhang et al. 2002, 2004) are found under Qualitative Analysis of Research Paper Studies and Quantitative Analysis.

QUALITATIVE ANALYSIS OF RESEARCH PAPER STUDIES

The main steps for freehand 3-D US imaging have been organized into the following subsections, which are accompanied by relevant references that focus more on freehand methods. It is noteworthy that except for the acquisition stage, all stages follow similar procedures for both freehand and sensorless methods.

Scanning B-slices and localization

To acquire a 3-D US volume, an operator must sweep a US probe on the ROI using couplant materials such as gels. The desired data are captured and sent to a computer for storage, and are shared with other clinical departments and finally analyzed by physicians.

Table 5. Grading system for assessment and exclusion of references

Grading criterion	Points	Acronym
3-D ultrasound reconstruction is performed without being limited to only a single medical application	1	S
Tracking system and acquisition data approach are explained completely	1	T
The results are clearly shown and discussed on 3-D volumes	1	R
Calibration method is demonstrated and phantom or specimen is discussed	1	C
Reconstruction algorithm is explained clearly	1	A
Literature is reviewed and a complete system of 3-D ultrasound reconstruction is depicted	1	L

Table 6. Numbers of references with each final grade

Grade	No. of references
0	14
1	24
2	48
3	68
4	123
5	82
6	12
Total	371

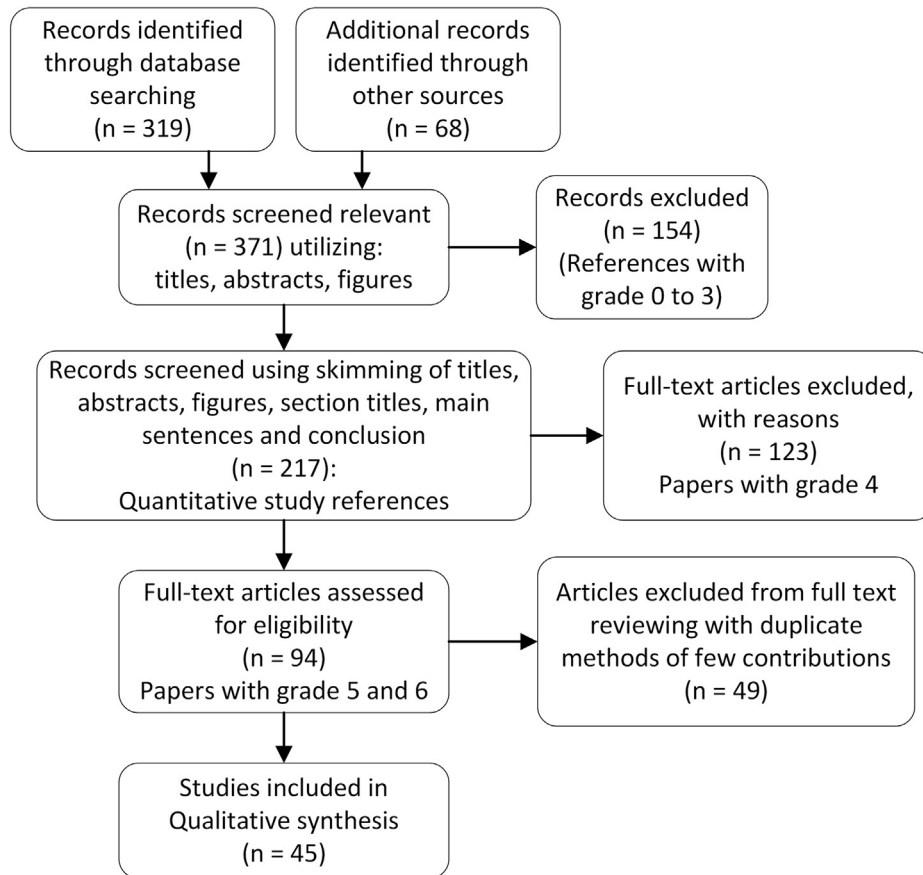


Fig. 4. Modified version of the PRISMA flow diagram consisting of reference exclusions and inclusions.

Some freehand methods include an interactive and real-time reconstruction system using an optical tracking device with which the operator can view the processes and rescan areas whenever the results were poorly covered on three orthogonal slice views of the 3-D US volume (Gobbi and Peters 2002; Ohbuchi et al. 1992; Welch et al. 2000).

A peripheral interface controller (PIC) has been used to control data flow and grabbing in a freehand 3-D US system, and this PIC module enables researchers to reverse the procedures and reach the quality of 2-D conventional images (Barry et al. 1997). They also could reach a better quality of B-slices by using images from multiple angles of insonation. The FOV of the US volume could be extended using a 3-D probe with an optical tracking system and a block-based registration of the ROI sections (Poon and Rohling 2006). An electromechanical system for use in freehand 3-D US imaging has been designed in which the motion of the 2-D US transducer is controlled through a hybrid (tilt and linear) motion trajectory automatically (Neshat et al. 2013). One part of the captured data is formed from fan-shaped motion, and another part is formed from linear motion; these

data are then combined to create a 3-D volume. The output result has the largest FOV from only one sweep of the transducer on the ROI after the 3-D US volume reconstruction.

Some researchers have focused on an autocorrelation function of the speckle signal and the decorrelation between a pair of images, to find the relationship between those images (Chen et al. 1997; Dickinson and Hill 1982; Li 1995; Tuthill et al. 1998; Wagner et al. 1987). These mathematical approaches made possible the following research on sensorless 3-D US reconstruction. The direction of the probe trajectory must be limited in many sensorless studies, which is a restriction, but recently, a sensorless reconstruction method has been designed using a regression-based distance measurement, interpolation techniques and unconstrained freehand data without any limitation on the trajectory (Housden et al. 2006b; Prager et al. 2003). A strategy for interpolation between the correlation peaks of the measured samples has enabled the tracking of simultaneous lateral and elevational motion of a transducer to be more accurate than in other similar methods (Housden et al. 2006a). The local transformation between a pair of B-scans and

backscattered echoes is calculated from a roughly ellipsoidal volume (resolution cell) around a pixel. This transformation was used to find a link between those pairs of B-scans and their out-of-plane offsets. Furthermore, a new technique was proposed for making out-of-plane distance measurements between the US images using an analysis of the speckle statistics (Hassenpflug et al. 2005; Prager et al. 2003).

In a proposed freehand method, one standard small camera is rigidly mounted on a US transducer. Data from the camera use skin features (Sun et al. 2013; Sun and Anthony 2012), or temporary tattoo stickers with rich features attached to the skin surface (Horvath et al. 2012; Rafii-Tari et al. 2011), while the probe was moving on the skin, were used to determine the probe position.

Detection of electromagnetic interference from electromagnetic tracking sensors was used for a freehand approach to estimate the probability density function of position and orientation measurement errors, and then, the 3-D US volume was constructed from multiple standard views using 3-D object-based registration and level set segmentation (Yu et al. 2005, 2006a) instead of a single view. As a tracking device for the freehand technique, a sliding bar with a linear positioning sensor that communicates with a computer wirelessly through a Bluetooth module was used to determine the linear location of the probe (Huang et al. 2013). To increase the accuracy of the location data by decreasing the interference and the ferromagnetic effect of the US probe and the tracking sensor, it is better to attach an electromagnetic sensor using a Perspex strip farther from the transducer (Barry et al. 1997).

A new spatial localizer was designed that was composed of flexible tape with several fiber-optic sensors along its length (Pagoulatos et al. 2000). The tape is attached to the probe, and the probe location data are found by detecting changes in the end of the tape with respect to its base using the light angles that reach the

fiber-optic sensors. This device is insensitive to the materials in the environment; it is not costly and has no occlusion problem. Figure 5 illustrates the main steps of the scanning and localization methods. Many scanning devices and tracking systems are listed in Table 1.

Acquisition

In freehand systems, it is vital to synchronize images and their corresponding locations using temporal calibration techniques. Furthermore, the tracking sensors send only the position and orientation of the probe itself instead of each pixel's location; thus, it is also important to find the location specification of each pixel of the B-scans and its corresponding voxel in the final volume. This relationship between pixels and voxels can be found by spatial calibration methods.

In the majority of calibration approaches, researchers have focused primarily on accuracy, whereas there are other factors, such as speed, robustness, automation, real-time feedback of precision, sterilization of the calibration phantom and easy assembly. Figure 6 illustrates some of the well-known phantom designs reported in the literature for visualization or calibration purposes. A real-time ultrasound calibration method with minimal human intervention was designed and tested (Chen et al. 2009). This technique was equipped by automatic error retrieval and accuracy control based on the known ground truth. Fiducial registration error (FRE) was used as feedback for the calibration accuracy along with fully automatic determination of the points in the images. The calibration method can speed up and requires only a single US image, which is captured from a custom-designed phantom (Pagoulatos et al. 2001). Similarly, it is possible to perform the calibration process in only a few minutes when a special phantom (the Cambridge phantom) is used (Prager et al. 1998b). By use of the Hough transform and a new robust estimator of the position, an automatic, robust, fast and reliable spatial calibration technique was

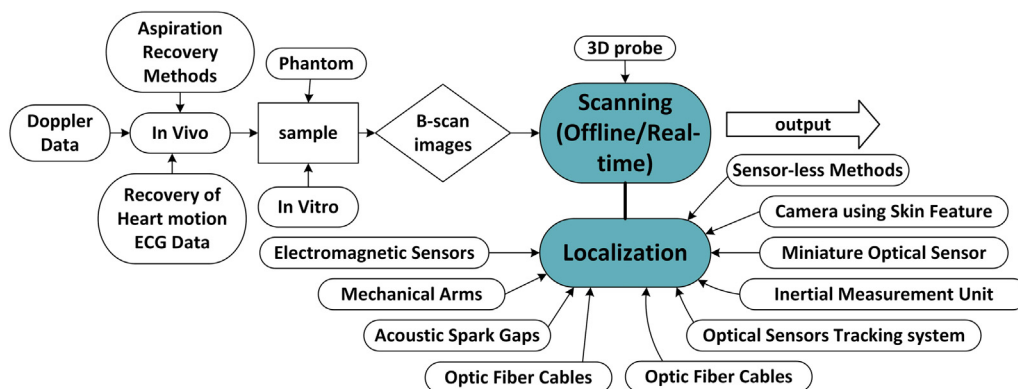


Fig. 5. Scanning processes and localization methods in 3-D US imaging systems.

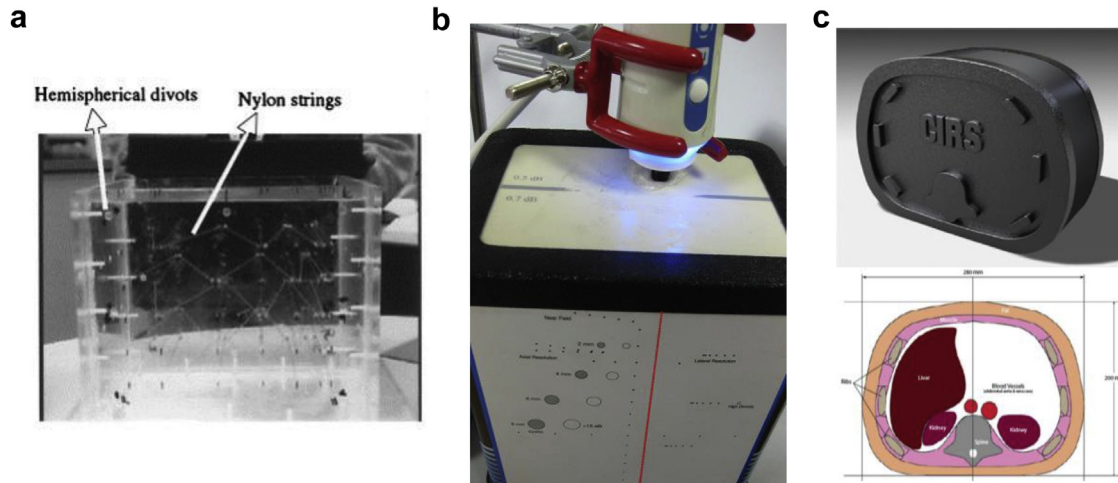


Fig. 6. Some of the phantom models used in 3-D ultrasound imaging. (a) Custom-designed calibration phantom with nylon strings (Pagoulatos et al. 2001). (b,c) Standard phantoms are used for calibration or visualization purposes, made by Computerized Imaging Reference Systems (CIRS) (Sasso et al. 2010).

designed using known 3-D geometric properties of the phantom (Rousseau et al. 2005).

Temporal calibration is performed by quickly moving the transducer away from the skin and making a step change in both the image content and location data (Prager et al. 1999). Another point to consider in the acquisition step is the US device output in terms of the data type, which can be an analogue signal and must be converted to digital with minimal information loss.

Location information that is acquired from the sensors is noisy. Therefore, some filtering methods must be employed to find the most corresponding position and orientation of each US B-scan among the location data (San José-Estépar et al. 2003). The general steps of the acquisition procedure are illustrated in Figure 7, and a list of phantoms that were used for calibration purposes is provided in Table 1. Many calibration techniques and positioning systems have been reviewed comprehensively in the literature (Dewi et al. 2015; Mercier et al. 2005; Poon and Rohling 2005; Tatar 2006).

Pre-processing

Before reconstruction of a 3-D volume from the acquired data, many enhancement methods, such as noise

filtering, cropping of redundant information and histogram equalization, are often applied to the B-scans to improve their quality. For example, median filters have been used for speckle reduction and edge preservation in B-slices (Huang and Zheng 2008).

Segmentation techniques in the 2-D domain are important for 3-D surface reconstruction because the segmentation result in 2-D is collected to form the 3-D surface and is also used for surface rendering (Treece et al. 2001a). In many studies, depending on the application, researchers tend to find a 3-D surface of the ROI using 2-D images instead of a 3-D volume reconstruction from the voxel data. In these cases, segmentation, registration and interpolation accuracy and speed are vital for producing better results. The usual pre-processing methods that are used in 3-D US imaging are depicted in Figure 8.

Reconstruction

The main part of 3-D US imaging, especially for freehand methods, is 3-D volume construction, which is conducted by the execution of interpolation and reconstruction algorithms. Many approaches have been designed and accomplished on the data to handle the

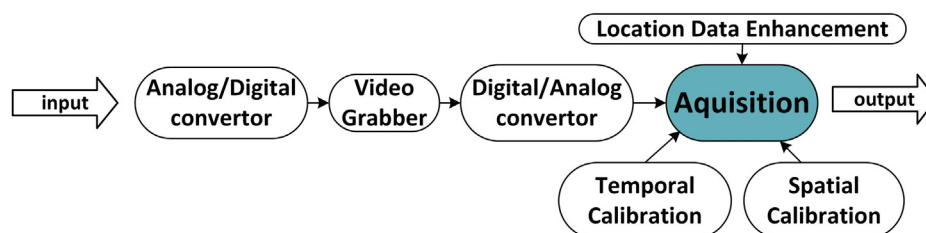


Fig. 7. Steps of the acquisition process.

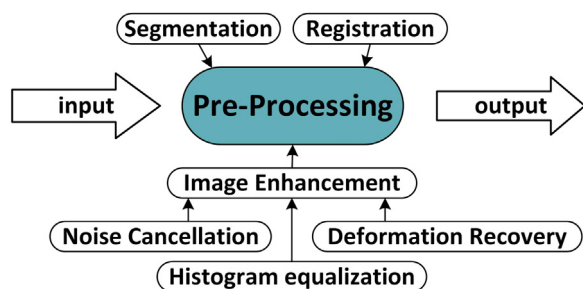


Fig. 8. Steps of the pre-processing stage.

translation of image pixels into voxels. Figure 9 is an overview of reconstruction methods. In general, the main purpose of these algorithms is to construct a 3-D US volume on a regular or irregular grid from a set of B-scans with minimum computational requirements and without damaging or losing the underlying shape of the data. Comparisons of these algorithms can be found in the literature (Miller et al. 2012; Rohling et al. 1999a; Solberg et al. 2007; Wen et al. 2015).

Selection of a suitable grid for the construction of the volume is one of the main issues in volume reconstruction. To have an appropriate 3-D grid, specifications of the coordinate system and the extent of the reconstruction volume must be defined in advance, especially when there is less information density around the outermost zones. In general, a desired volume grid has two characteristics (San José-Estépar et al. 2003): (i) a sufficient number of data samples, and (ii) a homogeneous sample density regardless of the distance to the borders.

To find the best regular grid for fitting the data, one approach is to use the bounding box method. Principle component analysis (PCA) is employed to select the best coordinate system, and an adaptive Gaussian kernel is used to find the optimum voxel size and reduce aliasing (San José-Estépar et al. 2003). In one research study (Roxborough and Nielson 2000), this problem was considered to be a trivariate scattered data approximation problem, and an arbitrary rectilinear box that was broken down into a tetrahedral grid was used as a domain.

Bayesian-based non-local interpolation using the gamma distribution along with the fast marching method and many denoising approaches, such as the Pearson distance function and filtering approaches, has been applied

to produce high-quality volume reconstruction in a free-hand method (Wen et al. 2015). A joint process of reconstruction and alignment was also used in a Bayesian framework, in a similar approach (Sanches and Marques 2003), and optimization algorithms were used to estimate the volume and alignment parameters in the 3-D volume construction.

Subpixel extension to phase correlation was used to track the points between subsequent frames for the volume reconstruction (Gilliam et al. 2006). This method exploits the well-known property of the Fourier shift, which is shifting in the spatial domain and corresponds to a linear phase shift in the Fourier domain.

Many interpolation methods have been proposed for volume reconstruction. For example, a 2-D non-rigid registration algorithm establishes a spatial correspondence between neighboring B-scans using B-splines (Rueckert et al. 1999), and is applied for the interpolation of B-scan pixels (Penney et al. 2004). A variational interpolation method for reconstructing the volume of interest (VOI) directly from contours segmented from the original freehand B-scans in the pre-processing step was proposed in a research study (Deng et al. 2011, 2012). Kernel regression can be applied as a two-step reconstruction algorithm that consists of bin-filling and finding a non-parametric estimation for all of the volume data from the previous sample of sparse data (Chen et al. 2014). An interpolation algorithm was proposed in a research study (Huang et al. 2015) that takes advantage of third-order Bezier curves for approximation of the voxels located in a control window with four control points. It is revealed that the distance-weighted interpolation can reduce the computation time in comparison with that for the Gaussian convolution kernel method. A simultaneous joint method that employs noise filtering, image interpolation and optimization was proposed for a volume reconstruction algorithm (Sanches and Marques 2000), in which the Rayleigh data distribution was used as a description of a multiplicative noise type that is observed in US data.

Comparison studies and two common classification methods for freehand volume reconstruction in addition to descriptions of other popular methods, such as distance-weighted interpolation, have been surveyed in detail in the literature (Rohling et al. 1999a; Solberg

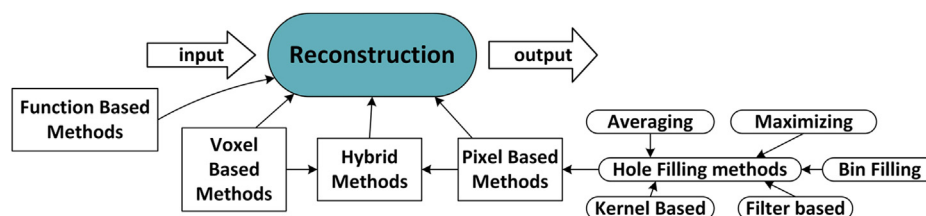


Fig. 9. A classification of reconstruction methods.

et al. 2007). In a study, the use of splines for reconstruction purposes was investigated (Rohling et al. 1999a), and for this reason, the radial basis function (RBF) approximation method was implemented and has been compared with other similar approaches. Median filters are applied to the topic for better reconstruction (Huang and Zheng 2008). An adaptive version of distance-weighted (DW) interpolation was employed for reconstruction interpolation in a freehand method, and this method could adaptively adjust the parameters related to the weight distribution in a squared distance-weighted (SDW) interpolation according to the ratio of the local variance and mean in a spherical neighborhood of each voxel (Huang and Zheng 2006a). In a similar approach, adaptive Gaussian distance-weighted (AGDW) interpolation was employed for volume reconstruction (Huang et al. 2009b).

In Gobbi and Peters (2002), the PNN method with different kernels accompanied by a boxcar function was applied to the data for volume reconstruction. For the interpolation, PTL (pixel trilinear) was applied with large kernels and radially symmetric splints. These methods are alternative ways of using RBFs with very computationally intensive, but very high-quality reconstructions. An improvement on the hole-filling technique in the PNN algorithm was proposed (Dewi et al. 2009) using an improved Olympic operation. This approach estimates empty voxel values by sorting the neighboring voxels and averaging them to attain a final value for the empty voxels. The fast marching method (FMM) was used in a study for freehand reconstruction by combining the voxel nearest neighbor (VNN), pixel nearest neighbor (PNN) and DW methods for interpolation (Wen et al. 2013).

Direct surface construction from data without interpolation of the voxel array was introduced in the literature by many researchers (Zhang et al. 2002, 2004). A set of semistructured ultrasound pixel data was considered as a single function to extract the surface without reconstructing the pixels into voxels. Radial basis function approximation was fitted to the group of data samples, and then, an explicit surface was obtained by iso-surfacing the function. Another 3-D US imaging system was proposed (Prager et al. 2002) that allows immediate visualization of the data without constructing a voxel array. Direct frame interpolation was designed in another research study with no additional hole filling for VOI (Scheipers et al. 2010). It is feasible because all of the voxels are defined by the arrangement of original and additionally constructed B-scans.

Stradx for freehand 3-D US imaging is one of the popular software systems for freehand 3-D US image reconstruction (Prager et al. 1999). This software was developed under the UK Engineering and Physical Sciences Research Council's Stradivarius project. Two cir-

cular buffers were used to select the most recent data within a specific age range; any arbitrary plane can be visualized in Stradx.

Ultrasound volume reconstruction can be interactive in realtime provided there is accurate feedback from the 3-D US system during acquisition with the MITK software (Dai et al. 2006). VOLUS (Varandas et al. 2004) and a new version of Stradx called Stradwin (MacGillivray et al. 2009) have also been designed for freehand 3-D US reconstruction. Many of the freehand reconstruction methods and their performances are listed in Table 1.

Visualization and volume rendering

Ultimately, diagnosticians and physicians like to see any arbitrary slices of the volume interactively without any visible gaps between the voxels. In many applications, visualization of the ROI surface is important to measure organ volume or any quantitative properties. For example, in obstetrics and gynecology, surface rendering of the fetus and its organs from the US volume is beneficial for the assessment of many prenatal diseases and antenatal treatments.

Three-dimensional rendering and visualization techniques were applied for US volume illustration. A method of 3-D surface visualization was presented using marching cube reconstruction and ray casting volume rendering (Kohyama et al. 2005). Narrowband and planar volume rendering was used for 3-D US rendering and visualization (Gee et al. 2002). Visualization techniques for 3-D US transducers are explained in the literature (Stetten and Tamburo 2001). For software used for visualization, refer to Table 1.

A freehand method for measuring the volume of large organs without interpolation of the voxel array was proposed (Treece et al. 2001a) in which only 10 to 20 selected B-slices were segmented. In many cases, to quantify the volume of an anatomic organ and its morphology, segmentation methods can be applied to the data (Fenster and Downey 2003) before the registration process. Figure 10 illustrates the common approaches in this step of freehand 3-D US imaging. GPU's enable parallel computation for reconstruction and rendering (Dai et al. 2010; Elnokrashy et al. 2009), and higher performance GPU-based methods are considered a promising technique for the deployment of realtime methods.

QUANTITATIVE STUDIES

Based on the 217 references graded ≥ 4 , we realized that researchers from the United States and the United Kingdom studied this subject more often than researchers from other countries (see Fig. 11). The numbers of

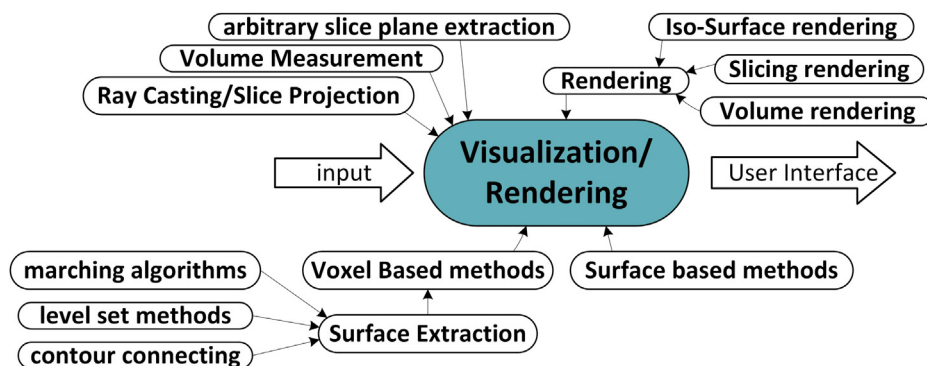


Fig. 10. Common 3-D US visualization and rendering techniques.

publications by researchers who published more than seven papers with a grade ≥ 4 are illustrated in Figure 12. From that figure, the names of five researchers whose names are repeated more than the others in the literature are listed in Table 7.

We can see from Table 7 and Figures 11 and 12 that researchers at the University of Cambridge are the leaders thus far (March 2017); they have published many research studies that provide valuable software (Stradx and Stradwin), major reviews of the methods in 3-D US imaging, a special type of phantom called the Cambridge phantom, many sensorless techniques and so forth. In Table 8, we listed the popular publications on the subject in order, starting with the highest number of published articles with grade 4 and more. The results revealed that *Ultrasound in Medicine and Biology* is the most popular journal, and the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) is the most popular conference for the topic.

Three-dimensional US imaging has been an interesting and demanding subject for three decades, as can be realized from Figure 13. We can also assert that between 1998 and 2009, more research contributions were published in the literature. Even though the use of 3-D ultrasound with 2-D arrays and internal and external mechanical fixtures has increased in recent years, the popularity of basic research on freehand 3-D US has been diminishing as we can see from the year of publication of five popular references in terms of citation (updated on July 2016) (see Table 9).

In terms of performance analysis and assessment, we understood from the literature that there are no standard criteria and system setup. For this reason, many researchers tend to express their results in terms of visual volumes or, eventually, in terms of volume estimation measurements or calibration accuracy. Table 1 indicates that there are few studies that compare their method with other similar techniques in the subject area. This lack of criteria is due to the differences between each

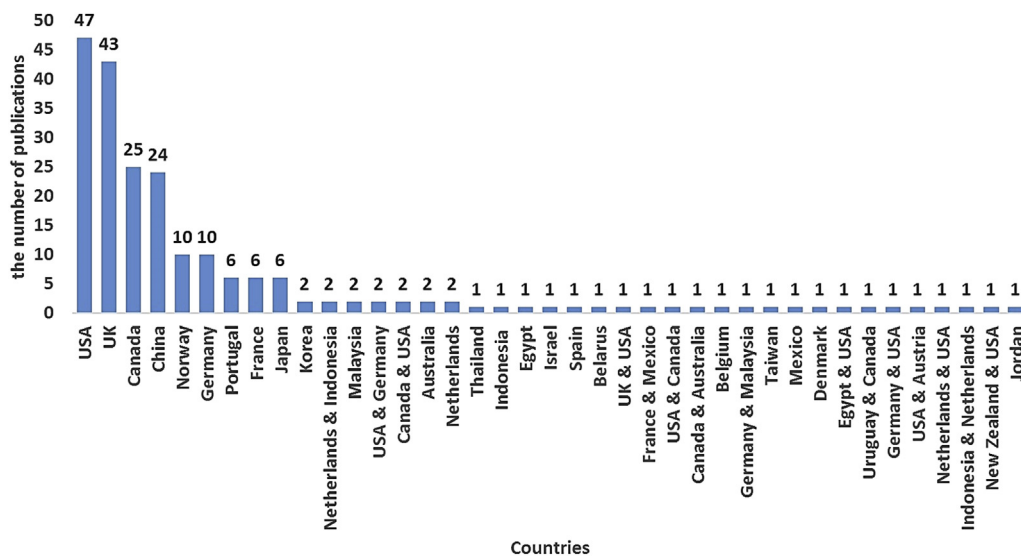


Fig. 11. Number of publications from each country.

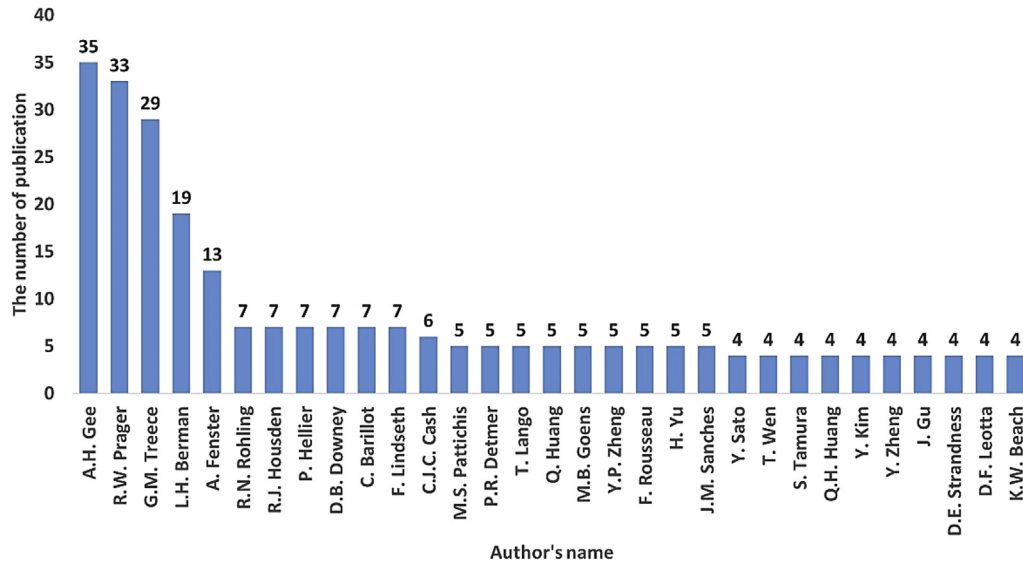


Fig. 12. Number of publications related to each author.

study and the focus point of each researcher. Differences between system setups, hardware implementations, calibration phantoms and visualization software make it impossible to determine which freehand reconstruction method is the best in terms of its performance, speed and reliability.

As Table 2 indicates, clinical applications of freehand systems can be easily classified into abdominal areas, such as kidney, liver and bladder; the vascular sys-

tems of the neck, head, legs and heart, such as the thyroid, spine, and scapular, carotid, coronary and scoliosis arteries; obstetrics and gynecology for prenatal diseases; and oncologic diseases, such as prostate, breast and brain cancers. Furthermore, needle path finding, laparoscopy, endoscopy, endosonography and endorectal and endoluminal ultrasound can be considered other main clinical applications of freehand systems.

DISCUSSION

Although many articles have been published on this topic, our review was limited to primary publications that describe the innovative techniques for freehand 3-D US

Table 7. Most repeated researcher names in the literature

Name of author	No. of publications	Author citation from Scopus database	Affiliation
Andrew H. Gee	35	2888 total citations by 1934 documents	Department of Engineering, University of Cambridge, Cambridge, UK
Richard W. Prager	33	2229 total citations by 1495 documents	Department of Engineering, University of Cambridge, Cambridge, UK
Graham M. Treece	29	1762 total citations by 1148 documents	Department of Engineering, University of Cambridge, Cambridge, UK
Laurence H. Berman	19	2435 total citations by 1878 documents	University of Cambridge, Cambridge, UK
Aaron Fenster	13	9756 total citations by 5927 documents	Department of Medical Biophysics, Western University, London, Canada

Table 8. Popular journals and conferences on the topic in the literature

Journal/conference with >4 references in our database	No. of references in our database
<i>Ultrasound in Medicine and Biology</i>	70
International Conference on Medical Image Computing and Computer-Assisted Intervention	20
<i>IEEE Transactions on Medical Imaging</i>	15
<i>Ultrasound in Obstetrics and Gynecology</i>	12
<i>Medical Image Analysis</i>	12
<i>IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control</i>	9
<i>Ultrasonics</i>	9
<i>SPIE Medical Imaging</i>	6
<i>Computerized Medical Imaging and Graphics</i>	4
<i>American Journal of Cardiology</i>	4
<i>Journal of Clinical Ultrasound</i>	4
<i>International Journal of Computer Assisted Radiology and Surgery</i>	4

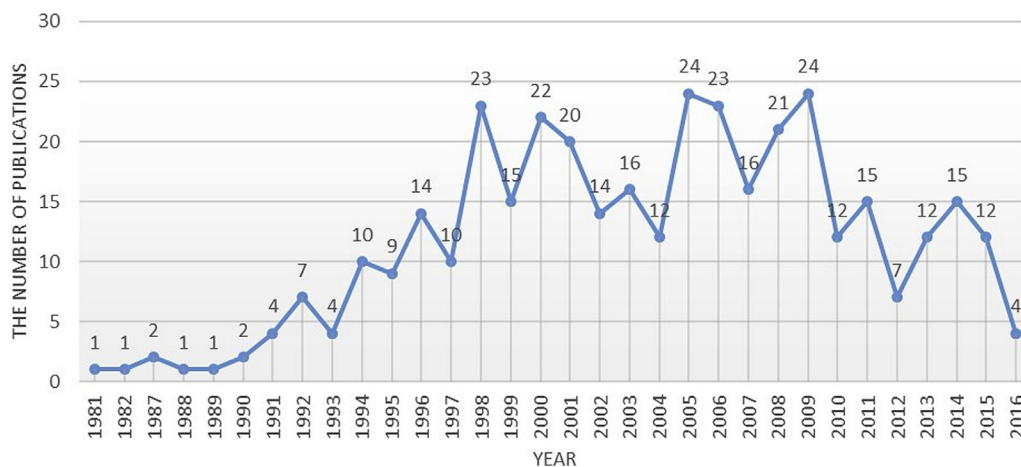


Fig. 13. Popularity of the topic among researchers from 1981 to 2016.

imaging and its applications in medicine. Our search terms were intentionally broad; we eliminated articles whose abstracts indicated the articles described only the results of an application, and we eliminated articles that focused on one process of the systems without explaining the other stages. Many dissertations and publications in other languages that address 3-D ultrasound imaging are not included in our review.

From the qualitative studies, we can assert that in freehand methods, reconstruction algorithms and spatial calibration have been the main research topics. In freehand systems, electromagnetic and optical sensors have been employed in many applications, whereas other types of position tracking remain prototypes. Research on 3-D Ultrasound imaging faces challenges in obtaining higher-quality 3-D US images:

- The quality of each B-scan must be acceptable for clinical application; otherwise, it should be improved using both hardware and software approaches. For example, making transducers with very small piezoelectric materials is a challenge for the electronics industry as there is always trade-off between size, noise, beamforming and image quality.
- The accuracy of the tracking system is an essential factor for the quality of volume reconstruction in freehand methods. Volume reconstruction from 2-D B-scans without a location tracking system is feasible with sensorless techniques, but at lower accuracy. There is a demand for the design of more accurate, robust and reliable tracking systems or sensorless methods.
- Spatial and temporal calibrations should be fast and accurate, especially in clinical applications. By use of a special phantom, spatial calibration can be performed very rapidly and accurately from only one B-scan.
- In some cases, especially real-time applications, reconstruction of the volume from B-scans can be skipped using reslicing methods, whereas one can see the desired slice of the ROI immediately from the acquisition process instead of viewing a 3-D volume.
- Higher accuracy in registration and segmentation in two and three dimensions is indispensable for comparing US images with other imaging modalities, surface rendering and field-of-view extensions. These are also important in the calibration stage for finding known specification structures in images, visualization procedures for surface rendering and quantitative

Table 9. Five popular articles in the literature in terms of citations

Researcher name	Year published	Article title	Publication	No. of citations
Aaron Fenster	2001	Three-dimensional ultrasound imaging	<i>Physics in Medicine and Biology</i>	525
Paul R. Detmer	1994	3-D ultrasonic image feature localization based on magnetic scan head tracking: <i>In vitro</i> calibration and validation	<i>Ultrasound in Medicine and Biology</i>	278
Laurence Mercier	2005	A review of calibration techniques for freehand 3-D ultrasound systems	<i>Ultrasound in Medicine and Biology</i>	265
Gronningsaeter	2000	Snowed, an ultrasound-based neuronavigational system	<i>Neurosurgery</i>	232
R. N. Rankin	1993	3-D sonographic reconstruction: techniques and diagnostic applications	<i>AJR American Journal of Roentgenology</i>	209

measurements of ROI. Although many state-of-the-art techniques of segmentation and registration have been proposed in the image processing literature, still few studies have used those methods in freehand 3-D US imaging.

Many problems must be addressed before freehand 3-D US imaging systems can be used as widely as the 2-D US imaging systems. Issues such as low accuracy of calibration techniques and tracking systems, low efficiency of reconstruction algorithms, non-standard phantom models, inherent artifacts in images of US apparatus and low-quality and noisy US images must still be solved. In addition, there are no standard criteria for comparing different freehand 3-D US imaging systems to check the advantages and disadvantages of each technique. Thus, in many studies, the results are illustrated without considering a standard criterion for results evaluation and comparison with other methods. This conclusion is obvious from [Table 1](#), which lists a large number of freehand methods without clear results or without any comparison studies in terms of performance or computational complexity.

After images are acquired using one medical imaging modality, regions of interest must be outlined using segmentation methods and compared with other images using registration approaches. These two methods are two of the most important challenges in medical applications not only for freehand 3-D Ultrasound, but for all other 2-D or 3-D modalities. For example, the accuracy of organ volume estimation highly depends greatly on image segmentation precision, and better segmentation means better volume measurement. Because of the different characteristics of images obtained with different imaging modalities and the different kinds of organs and their postures, we cannot apply one general segmentation and registration technique to all images. Thus, depending on the image modality and ROI, many segmentation and registration techniques have been proposed and research in this area remains active.

From our quantitative studies, we learned that the number of centralized study groups that focused on this area is limited to laboratories in the United States and United Kingdom. Furthermore, fewer contributions have been published in the literature in recent years compared with the previous decade. We surmise that this finding either reflects a decrease in general interest in freehand ultrasound or indicates a change in the direction of the studies, for example, to a focus on improvements in hardware rather than software, such as beam-forming techniques for 3-D US probes or special transducer designs for 3-D US imaging. Our quantitative studies about the leading researchers and research groups

correlates with the results of a study in the literature ([Boctor et al. 2001](#)).

Great effort has been extended to improve each section of the freehand 3-D US imaging systems, such as different acquisition techniques, calibration methods and reconstruction algorithms. Nevertheless, few techniques have been found for visualization either offline or in real time. Different artificial intelligence techniques, such as classification, clustering, fuzzy logic, heuristic optimization techniques, and other methods from machine learning and computer vision, can be useful in the reconstruction part, but continue to be disregarded. In real-time approaches, many researchers have only displayed their prototypes without using them in clinical applications. Super-resolution imaging algorithms and similar techniques using the Doppler effect are in progress. One recent, interesting subject in the literature is the development of fusion images from the 3-D US volumes with other imaging modalities such as fusion 3-D US and MRI imaging for prostate biopsy ([Baust et al. 2013](#); [Hu et al. 2017](#); [Natarajan et al. 2011](#); [Sierra et al. 2015](#); [Xu et al. 2008](#)), which could provide physicians with valuable information for better diagnosis and treatment. Many modern commercial US systems have been used with 3-D probes, but they are usually manufactured with closed-source platforms, which is an obstacle to improving 3-D US imaging in academia.

CONCLUSIONS

A systematic review of 3-D ultrasound imaging methods was conducted to determine the prevalence and trends in research on 3-D US imaging and volume reconstruction between 1970 and 2017. We investigated the approaches that have been used to date, the subtopics that are more popular in the literature and the leaders, and we also discussed the advantages and disadvantages of current technologies.

We understand from this study that freehand systems are more prevalent in the academic environment, whereas in clinical applications and industrial research departments, most studies have focused on 3-D US transducers, its applications, improvement of hardware performance and review of relevant studies. If the industrial and academic realms combine in providing open-source materials, evaluation criteria and gold standards, there would be a significant increase in the speed of development and research on 3-D US imaging and freehand methods. Although researchers, especially in the United States and United Kingdom, have contributed greatly to the literature, in recent years, there has been a decrease in general interest of researchers in freehand ultrasound.

Ultimately, designing a comprehensive and robust method for capturing 3-D volumes with less difficulty,

higher precision and higher speed has become partly feasible in recent years because of advances in science and technology, such as high-performance central and graphics processing units, efficient and better calibration methods and reconstruction and visualization algorithms and software, cheaper, smaller and portable US devices, and significant growth in knowledge about acoustic sound. Many studies have proposed methods for 3-D US imaging using the freehand approach, and freehand ultrasound is still an interesting active area for researchers, although many unsolved problems remain to be addressed.

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